

Homework 5

BSTA 551

Directions

For simulation problems, set your seed to `set.seed(551)` for reproducibility.

You must show all of your work to receive credit.

This homework covers material from **Lessons 12–14**: Introduction to Hypothesis Testing, Tests About a Population Mean and Proportion, and P-values, Power, and Sample Size Determination.

Questions

Problem 1: Hypothesis Testing Framework (Lesson 12, Devore 9.1)

A hospital is evaluating a new blood pressure monitoring device. Based on a large historical dataset, resting systolic blood pressure in healthy adults follows a Normal distribution. The current standard monitoring protocol has a known mean of $\mu_0 = 120$ mmHg with known standard deviation $\sigma = 15$ mmHg. Clinicians suspect the new device reads systematically **higher** than the true value.

A validation study measures $n = 36$ healthy adults with the new device and obtains a sample mean of $\bar{x} = 124.5$ mmHg.

(a) State the null and alternative hypotheses formally. Is this a one-sided or two-sided test? Justify your choice based on the clinical context.

(b) The test statistic is $Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$. State the distribution of Z under H_0 and compute its observed value z_{obs} .

(c) Using a significance level of $\alpha = 0.05$, find the rejection region. Does the test reject H_0 ?

(d) Define **Type I error** and **Type II error** in the context of this specific problem (not just in abstract terms). What is the Type I error rate for this test?

(e) The hospital board argues that falsely concluding the device is biased when it is actually accurate would be very costly (they would discard the device). With this in mind, would you recommend using $\alpha = 0.01$ or $\alpha = 0.10$? Explain briefly.

Problem 2: One-Sample t Test (Lesson 13, Devore 9.2)

Researchers are investigating whether a dietary intervention reduces fasting glucose levels in pre-diabetic patients. Before the intervention, fasting glucose follows a Normal distribution with mean $\mu_0 = 105$ mg/dL in this population. A random sample of $n = 15$ patients who completed the 12-week program has the following fasting glucose levels (mg/dL) measured at the end of the study:

$$X = \{98, 103, 91, 107, 96, 88, 101, 95, 99, 102, 94, 97, 89, 100, 93\}$$

The population standard deviation is **unknown**.

(a) Conduct a hypothesis test at the $\alpha = 0.05$ significance level to determine whether there is evidence that the intervention **reduces** mean fasting glucose below 105 mg/dL. State hypotheses, compute the test statistic, find the critical value or rejection region, and state your conclusion.

(b) Compute the observed p-value. Interpret it in plain language (i.e., what does this number mean in the context of the study?).

(c) Compute a 95% one-sided confidence interval (lower bound, ∞) for the mean fasting glucose μ and verify that it is consistent with your conclusion from part (a). (*Hint: For a one-sided upper confidence bound/lower confidence bound, the critical value is $t_{\alpha, n-1}$, not $t_{\alpha/2, n-1}$.*)

(d) In R, use `t.test()` to verify your calculations. Show your code.

```
glucose <- c(98, 103, 91, 107, 96, 88, 101, 95, 99, 102, 94, 97, 89, 100, 93)

# Your code here
```

Problem 3: Test for a Proportion (Lesson 13, Devore 9.3)

A large multicenter clinical trial reports that the 30-day readmission rate for patients with heart failure treated under the standard care protocol is $p_0 = 0.22$ (22%). A new care management program is piloted at a regional hospital. Of the first $n = 180$ heart failure patients discharged under the new program, 31 were readmitted within 30 days.

(a) The hospital wants to know if the new program has a **different** readmission rate from the standard. State the null and alternative hypotheses.

(b) Compute the large-sample z test statistic:

$$Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$$

(c) Compute the two-sided p-value and state your conclusion at $\alpha = 0.05$.

(d) Compute a 95% confidence interval for p using the Wilson score interval (or the Wald interval). Is $p_0 = 0.22$ inside the interval? How does this relate to your conclusion in part (c)?

(e) A skeptic notes that $n = 180$ is not that large and questions whether the Normal approximation is valid. State the common rule-of-thumb for when the large-sample z test for proportions is appropriate, and check it for this problem.

Problem 4: P-Values, Power, and Sample Size (Lesson 14, Devore 9.2, 9.4)

Hospitals implementing **Enhanced Recovery After Surgery (ERAS)** protocols aim to reduce post-operative length of stay (LOS). Based on retrospective data at a large academic medical center, LOS for patients undergoing elective colorectal surgery under the standard protocol follows a Normal distribution with mean $\mu_0 = 7$ days and standard deviation $\sigma = 2.5$ days (treat as known for planning purposes). A research team wants to test whether ERAS reduces mean LOS, and would like to detect a clinically meaningful reduction of 1.5 days (i.e., $\mu_a = 5.5$ days).

(a) State the null and alternative hypotheses. Is this a one-sided or two-sided test? Justify your choice.

(b) For a **one-sided lower-tailed** z test with $\alpha = 0.05$ and a planned sample size of $n = 40$ patients, compute the power of the test against the alternative $\mu_a = 5.5$ days. Use the formula:

$$\text{Power} = \Phi\left(\frac{\mu_0 - \mu_a}{\sigma/\sqrt{n}} - z_\alpha\right)$$

where $z_\alpha = z_{0.05}$ is the upper α critical value. Show your work.

(c) The team sets a target of 85% power. Use `power.t.test()` in R to find the required sample size, treating this as a one-sample t test (the normal approximation is close for planning purposes). Use `sd = 2.5`, `delta = 1.5`, `sig.level = 0.05`, `power = 0.85`, `alternative = "one.sided"`.

```
power.t.test(
  delta = ___,
  sd = ___,
  sig.level = ___,
  power = ___,
  type = "one.sample",
  alternative = "one.sided"
)
```

(d) Create a **power curve** in R: for the one-sided lower-tailed z test, plot power as a function of n (from 5 to 120) fixing $\mu_a = 5.5$, $\sigma = 2.5$, $\alpha = 0.05$. Add a horizontal reference line at 85% power and a vertical reference line at the sample size found in part (c).

```
n_vals <- 5:120
mu0 <- 7; mu_a <- 5.5; sigma <- 2.5; alpha <- 0.05

power_vals <- pnorm(___ - qnorm(1 - alpha))

tibble(n = n_vals, power = power_vals) |>
  ggplot(aes(x = n, y = power)) +
  geom_line(color = "steelblue", linewidth = 1.2) +
  geom_hline(yintercept = 0.85, linetype = "dashed", color = "firebrick") +
  geom_vline(xintercept = ___, linetype = "dashed", color = "darkgreen") +
  labs(
    x = "Sample size (n)",
    y = "Power",
    title = bquote("Power curve: ERAS post-surgical LOS study"),
    subtitle = bquote(H[a]*":"~mu == .(mu_a)~"days,"~sigma == .(sigma)~","~alpha == .(alpha)
  ) +
  scale_y_continuous(limits = c(0, 1), labels = scales::percent)
```

(e) The ERAS study is ultimately run with $n = 55$ patients and produces a sample mean LOS of $\bar{x} = 6.1$ days.

- Compute the p-value for the one-sided lower-tailed z test.
- Write a one-sentence interpretation of the p-value suitable for a clinical audience.
- Is the result statistically significant at $\alpha = 0.05$? Is a reduction from 7 to 6.1 days clinically meaningful given the original target of 1.5 days? Discuss briefly — are statistical and clinical significance the same thing here?

Problem 5: Multiple Testing (Lesson 14)

A genomics study screens $m = 500$ genes for differential expression between cancer tissue and healthy tissue. After performing 500 independent hypothesis tests (one per gene) at level $\alpha = 0.05$, 37 genes are declared significant.

(a) Assuming all 500 null hypotheses are true, how many false positives (Type I errors) would you expect? What is this called, and why is it a problem?

(b) Apply the **Bonferroni correction**: what adjusted significance threshold α^* should be used for each individual test to control the family-wise error rate (FWER) at 0.05?

(c) Of the 500 tests, the 37 significant raw p-values are given below. Apply the Bonferroni correction and report how many genes remain significant.

```
set.seed(551)
# Simulate 37 "significant" raw p-values
sig_pvals <- runif(37, min = 0, max = 0.05)
sig_pvals
```

```
[1] 0.039991622 0.009029548 0.029696850 0.010112608 0.036451590 0.004385602
[7] 0.003395444 0.029779669 0.016211141 0.013747465 0.039203407 0.004591254
[13] 0.006939440 0.010781811 0.004885578 0.018662517 0.021031488 0.032292032
[19] 0.045465993 0.037860008 0.037340248 0.036234907 0.026430696 0.010631198
[25] 0.047648861 0.024383201 0.004109782 0.015042967 0.048505777 0.005563640
[31] 0.044336905 0.048007904 0.014847766 0.039496309 0.043135550 0.020120665
[37] 0.040028855
```

```
# Apply Bonferroni correction -- your code here
```

(d) Explain briefly why the Benjamini-Hochberg (BH) procedure for controlling the False Discovery Rate (FDR) is often preferred over Bonferroni in “omics” studies.